

Hospital Dashboard

Data Source:

<https://docs.google.com/spreadsheets/d/1UozE4prNlu86DbhpTgjY0u9uR-i-WH5Z/edit?usp=sharing&ouid=106658957114394760293&rtpof=true&sd=true>

Understand the Raw Dataset

The sheet contains transactional hospital visit data.

Column	Meaning
Visit_ID	Unique visit identifier
Visit_Date	Date of hospital visit
Patient_ID	Unique patient
Patient_Age	Age
Patient_Gender	Gender
Doctor_ID	Doctor
Department	Medical department
Diagnosis	Diagnosis type
Procedure	Procedure performed
Hospital_Name	Hospital
Payment_Method	Cash / Insurance / Card
Insurance_Provider	Insurance company
Procedure_Cost	Procedure charges
Medicine_Cost	Medicine charges
Room_Charges	Room charges
Total_Bill	Total amount

Fact_Hospital_Visit

Column
Visit_ID
Date_Key
Patient_ID
Doctor_ID
Procedure_ID
Diagnosis_ID
Department_ID
Payment_ID
Hospital_ID
Procedure_Cost
Medicine_Cost
Room_Charges
Total_Bill

Dim_Patient

Column
Patient_ID
Gender
Age
Age_Group

Formula of Age Groups:

if [Patient_Age] <= 12 then "Child"

else if [Patient_Age] <= 19 then "Teen"

else if [Patient_Age] <= 35 then "Young Adult"

else if [Patient_Age] <= 60 then "Adult"

else "Senior"

Dim_Doctor

Column
Doctor_ID
Department_ID
Doctor_Name
Specialty

Doctor_Name >> “Doctor” & Doctor_ID

Specialty >> Department

Dim_Department

Column
Department_ID
Department_Name

Dim_Date

Column
Date_Key
Date
Day
Month
Quarter
Year
Week

Dim_Procedure

Column
Procedure_ID
Procedure_Name
Procedure_Type

Procedure_Type

if [Procedure] = "Consultation" then "Consultation"

else if [Procedure] = "Blood Test" then "Lab"

else if [Procedure] = "X-Ray" or [Procedure] = "MRI" or [Procedure] = "CT Scan" or [Procedure] = "Ultrasound" then "Imaging"

else if [Procedure] = "Surgery" then "Surgical"

else if [Procedure] = "Physiotherapy" then "Therapy"

else "Other"

Dim_Diagnosis

Column
Diagnosis_ID
Diagnosis_Name
Disease_Category

Disease_Category

if [Diagnosis] = "Hypertension" or [Diagnosis] = "Diabetes" then "Chronic"

else if [Diagnosis] = "Fracture" or [Diagnosis] = "Migraine" then "Acute"

else if [Diagnosis] = "Asthma" then "Respiratory"

else if [Diagnosis] = "Infection" then "Infectious"

else if [Diagnosis] = "Cancer Screening" then "Screening"

else if [Diagnosis] = "Skin Allergy" then "Dermatology"

else "Other"

Dim_Payment

Column
Payment_ID
Payment_Method
Insurance_Provider

Dim_Hospital

Column
Hospital_ID
Hospital_Name
City
Region

City

if [Hospital_Name] = "Central Hospital" then "Riyadh"

else if [Hospital_Name] = "City Medical Center" then "Jeddah"

else if [Hospital_Name] = "Al Noor Hospital" then "Dammam"

else "Unknown"

Region

if [City] = "Riyadh" then "Central"

else if [City] = "Jeddah" then "Western"

else if [City] = "Dammam" then "Eastern"

else "Unknown"

model relationships in Power BI

After loading all queries, create these relationships:

- Fact_Hospital_Visit[Date_Key] → Dim_Date[Date_Key]
- Fact_Hospital_Visit[Patient_ID] → Dim_Patient[Patient_ID]
- Fact_Hospital_Visit[Doctor_ID] → Dim_Doctor[Doctor_ID]
- Fact_Hospital_Visit[Department_ID] →
Dim_Department[Department_ID]
- Fact_Hospital_Visit[Diagnosis_ID] → Dim_Diagnosis[Diagnosis_ID]
- Fact_Hospital_Visit[Procedure_ID] → Dim_Procedure[Procedure_ID]
- Fact_Hospital_Visit[Payment_ID] → Dim_Payment[Payment_ID]
- Fact_Hospital_Visit[Hospital_ID] → Dim_Hospital[Hospital_ID]

Important Healthcare KPIs

- Total Revenue >> Total_Bill
- Average Bill Per Visit
- Medicine % >> Medicine_Cost / Total_Bill
- Total Visits
- Distinct Patients
- Visits Per Patient

Power BI Dashboard Layout

Page 1 — Executive Overview

Cards:

- Total Revenue
- Total Visits
- Distinct Patients
- Avg Bill

Charts:

- Revenue by Department
- Visits by Month
- Visits by Hospital

Page 2 — Medical Analysis

Charts:

- Top Diagnoses
- Top Procedures
- Department Workload

Page 3 — Financial Insights

Charts:

- Revenue by Payment Method
- Revenue by Insurance Provider
- Cost Breakdown